

- Automated pipelines and reusable workflows
 - Tool integration and workflow consistency
 - Design comparison and team collaboration
 - Automatic simulations
 - Reliable RF with faster deployment
- They can enhance the performance of satellite systems, radar and air defence systems, IoT, and drones.

Figure 2 depicts headless RF engineering that integrates the automated Hardware-in-the-Loop (HIL) testbed for radio network and antenna analytics. A cloud managed software defined radio (SDR) platform works without any manual intervention.

The integration of AI and platform engineering in radio networks leads to faster testing, higher accuracy, rapid analysis, better throughput, stable and faster designs, improved performance, optimal coverage, reduced errors, easy deployment, faster innovation, and rapid rollouts.

Simulation on a 4nec2 platform for antenna and radio analytics

URL: <https://www.qsl.net/4nec2/>

The free and open source platform 4nec2 can be used for simulating antenna and radio analytics.

An open source platform like 4nec2 along with platform engineering can automatically execute antenna designs using:

- 3D radiation patterns
- VSWR vs frequency
- Gain patterns (polar plot)
- Impedance
- Smith chart

The effectiveness and performance of an AI and platform engineering-based radio network workflow can be elevated to a large extent with automated and reusable modules.

The platform engineering pipeline is:

- Electromagnetic RF 4NEC2 simulation
- Extraction of gain
- Extraction and analytics of Voltage



Figure 2: AI-driven automated radio networks and antennas using platform engineering

Table 3: Uses of integrating AI with platform engineering

Application	Platform engineering role	AI role
Antenna design	Workflow automation	Shape optimization
SDR receiver	Container deployment	Signal classification
RF simulation	Auto pipeline	Result prediction
Beamforming	Shared platform	Pattern learning
Spectrum sensing	CI/CD pipeline	Signal detection
MIMO systems	Resource sharing	Channel prediction
Patch antenna	Version control	Parameter tuning
Dipole design	Template reuse	Length optimization
Yagi design	Batch execution	Gain prediction
Radar systems	Automated testing	Target recognition
Satellite links	Cloud platform	Link prediction
EMI analysis	Auto reports	Noise detection
RF filters	Standard environment	Parameter search
IoT antennas	Shared libraries	Energy prediction
SDR transmitter	Container images	Waveform selection
Radio networks	Platform scaling	Traffic prediction
Antenna arrays	Pipeline reuse	Beam optimization
RF research	Knowledge sharing	Data mining
5G planning	Workflow templates	Site prediction
Defence systems	Secure platform	Threat detection